

Task 14 – Time Series Analysis (AirPassengers Dataset)

Objective:

Analyze AirPassengers time series data by performing decomposition, stationarity testing and forecasting using ARIMA.

Dataset:

AirPassengers dataset contains monthly totals of international airline passengers from 1949 to 1960.

Step 1 – Data Visualization:

The time series plot shows a strong upward trend and clear seasonality pattern, indicating non-stationary data.

Step 2 – Decomposition:

Seasonal decomposition separates the data into Trend, Seasonal and Residual components. The trend shows long-term growth and the seasonal component shows yearly repetition.

Step 3 – Stationarity Test (ADF):

ADF test checks whether data is stationary. Initial p-value > 0.05 indicated non-stationary data. Differencing was applied to make the data stationary.

Step 4 – ARIMA Model:

ARIMA(1,1,1) model was used. AR uses past values, I removes trend using differencing, and MA uses past errors.

Step 5 – Forecasting:

The model successfully forecasted the next 24 months of airline passengers, showing increasing future trends.

Conclusion:

Time series analysis successfully identified trend and seasonality. ARIMA model provided reliable future predictions.